DossierBench: Scoring a Recommender's Model of You on Calibration, Not Recall

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Abstract

Recommendation apps increasingly keep a standing preference profile of the user, built from conversation. The memory systems that build such profiles are evaluated on *recall*, which rewards committing and cannot see the failure that matters once a profile drives recommendations: asserting a preference the user never expressed. We introduce **DossierBench**, which scores a profile on *calibration*: does the system commit exactly where the conversation has given it grounds to? Each synthetic subject is a persona whose dining and going-out preferences are fixed on a 24-field card and surface gradually over a scripted conversation of many sessions. An *evidence map* records, for every field at every checkpoint, what the conversation so far has shown: STRONG, WEAK, or UNEVIDENCED. A system is served the conversation one checkpoint at a time and emits the whole card, abstaining where it will not commit. Committing to an unevidenced field is a fabrication even when the value is right; abstaining there is free. Fidelity over the timeline is normalized to a skill score with always-abstain at 0 and an oracle at 1. On 50 subjects (612 sessions) and four frontier backbones, two single-pass extractors that read the conversation directly calibrate well, while a stateful memory layer, evaluated on one backbone, scores below both at twice the call budget. We release the corpus, scorer, and reference systems; the benchmark scores the profile alone, before any recommendation is made.

1 Introduction

An app that recommends where to eat, drink, or go out is only as good as its model of the user it serves. Increasingly that model is a standing *preference profile* assembled from conversation: the user talks to the app over months, and the app maintains what it believes about their cuisines, budget, noise tolerance, travel radius, and usual company. The recommender conditions on that profile, so a wrong entry is not an isolated mistake but a standing bias on everything the app suggests next. The memory systems that build such profiles have largely been optimized for *recall*: given a fact stated earlier, can the system retrieve it later [Packer et al., 2023, Zhong et al., 2024, Chhikara et al., 2025]. Recall is necessary, but it is the wrong top-level target here because it prices only one of the two errors a profile can make. A profile that forgets a stated allergy recommends a dangerous dish; a profile that *invents* a preference the user never voiced also misleads the recommender, and a recall metric cannot see it. A system that commits a plausible value to every field can score well on recall while confidently telling the recommender that a user who never mentioned wine wants an alcohol-free venue, or that a user likes Thai food because they once booked a Thai restaurant for someone else.

We argue that the property to measure is **calibration under evidence**: a profile should commit to a field exactly when the conversation supports it, and abstain otherwise. We use *calibration* in this sense, whether the decision to commit matches the evidence, not in the probabilistic sense of stated confidence matching accuracy. The distinction is operational, not just epistemic. A field the profile

40th Conference on Neural Information Processing Systems (NeurIPS 2026). Workshop: Personalized, Aligned, Long-Term Memory for AI Systems (PALM) @ NeurIPS 2026.

leaves empty is one the recommender can handle (fall back to a popularity prior, widen the candidate set, or ask), whereas a fabricated field is one it will silently condition on, with nothing downstream able to tell the two apart. Making that checkable requires knowing, at each point in a dialogue history, what a careful reader *could* know. That is the central artifact of this paper: for each subject in the dataset we do not merely store the ground-truth persona, we derive, from the *evidence schedule* the conversations were written from (which session reveals which field, and how clearly), an *evidence map* grading every field at every checkpoint as STRONG, WEAK, or UNEVIDENCED. The map turns "the system abstained here" into a scoreable event: abstaining on an unevidenced field is correct, abstaining on a strong field is a miss, and committing to an unevidenced field is a fabrication whether or not the guess is right. We evaluate the profile in isolation: no recommendations are produced or ranked, and pricing each error by the recommendation it spoils is the next step (§8).

This paper makes the following contributions:

- **An evidence-conditioned benchmark for preference profiles** (§3), spanning 24 fields (20 preferences and 4 hard constraints) across 8 venue-recommendation groups. Each preference field is scored with a type-appropriate metric (set overlap, ordinal distance, or distribution divergence), and each hard constraint by whether any item is missed or invented. The harness serves one checkpoint of new dialogue at a time, without re-serving past prefixes or revealing the answer key. The 50-subject corpus is *synthetic*: sampled personas and evidence schedules, LLM-authored sessions, and an answer key derived from the schedule. The dataset and code are released.[1]
- **A schedule-derived, validated answer key** (§5): the evidence map is computed from what the schedule placed; the LLM-reader credit and hand corrections layered on top are small, constrained, and recorded, and every map is checked to be monotone.
- **A calibration score with fixed anchors** (§6): a fidelity trajectory, its area under the curve (AUFC), and a skill normalization pinned so that always-abstain $= 0$ and the oracle $= 1$; negative skill means *worse than emitting nothing*.
- **Reference measurements** (§7): three simple reference systems, two single-pass extractors on four frontier backbones (GPT-5.6 Luna, Gemini 3.7 Flash, DeepSeek V4 Pro, GLM-5.3 Flash) and one stateful memory layer on GPT-5.6 Luna. Both single-pass extractors clear the abstain floor on every backbone; the stateful layer falls below both on its backbone at twice the call budget. A transcript-grounded failure-mode reading (§7.1) traces what each system gets wrong and why.

2 Related Work

Conversational recommendation and preference elicitation. Learning a user's preferences through dialogue has a long history in recommendation. The system decides what to ask about next. It may ask for an item to rate [Christakopoulou et al., 2016] or an attribute [Sun and Zhang, 2018, Lei et al., 2020], and it recommends once it judges it has learned enough. In that setting an unmentioned attribute is something to ask about. In region-based elicitation it is an uncertainty set over the user's embedding to shrink [Nguyen et al., 2024]. It is never a value the system is scored for asserting. An LLM-built profile for recommendation simulation is scored by the downstream behaviour it produces [Wanyan et al., 2026]. That counts a correct-but-unsupported trait as a success. τ-Rec [Narasimhan and Narasimhan, 2026] is the nearest protocol. Its reveal tags (`volunteer` / `on_ask` / `hidden`) fix when each constraint is disclosed. It measures a roughly fourfold drop in success from volunteered to hidden constraints. But the tags steer only the user simulator and never enter the reward. The scored artifact is the recommended item, so an agent that *guesses* a hidden constraint scores exactly as one that learned it from the user's rejections. Ours is the belief-side instrument to that outcome-side one.

Dialogue state tracking. The nearest experimental instrument to ours is dialogue state tracking (DST), which tracks typed slots across turns, treating unmentioned slots as absent or explicit `none` values [Rastogi et al., 2020, Aksu and Chen, 2024]. The difference is what the gold state holds. DST tracks what the user said; we also charge a value that is true of the user when the transcript has not yet licensed it. DST also cannot represent a hint that narrows a field without pinning it. Two things follow that have no DST counterpart. One is a three-way evidence band instead of mentioned-or-not. The other is abstention as a scored positive move instead of the free non-event a correct `none` is. We do inherit two metric lessons. Slot accuracy is inflated by the mass of correctly empty slots [Kim et al., 2022], so our fidelity denominator is fixed to the evidenced fields. And a per-turn metric re-charges

[1] Data: `https://huggingface.co/datasets/tangents/dossierbench`; code (scorer, reference systems, prompts): `https://github.com/Hey-L-AI/DossierBench`.

one early error at every later turn [Aksu and Chen, 2024]. Our AUFC does this deliberately, since a memory holding a wrong value is wrong each time it is read.

Memory architectures and long-conversation benchmarks. MemGPT [Packer et al., 2023] manages a tiered memory with explicit paging. MemoryBank [Zhong et al., 2024] adds forgetting curves. Mem0 [Chhikara et al., 2025] extracts salient facts and consolidates them into a bank. These systems are evaluated by retrieval-style question answering over multi-session dialogue [Maharana et al., 2024, Xu et al., 2022] or long documents. We treat them as *systems under test* and ask a different question: not "can you retrieve the fact" but "do you know when you don't know." While LoCoMo's adversarial slice credits declining to answer, answerability is treated as a static per-question label rather than an evolving function of accumulated evidence. Furthermore, on answerable questions, matching responses receive full credit regardless of whether they were soundly inferred or guessed. No such benchmark grades a claim against the evidence state at the time it was made. So none separates a correct abstention from a lucky guess, and none prices a fabrication that happens to be right.

Evidence-conditioned scoring of inferred profiles. The failure we target is asserting a preference the conversation does not support. It has only recently been measured directly. MirageBench [Sun et al., 2026] comes closest. It scores inferred user-profile claims on *evidential support rather than truth*, and it labels a claim that goes beyond the revealed evidence as over-inference "regardless of whether ... [the profile] agrees with it." Its authors call for stored user information to be tagged with its epistemic status, which is the three-way banding we compute. It differs from our setting in exactly the ways that motivate ours. Its evidence is a fixed handful of revealed facts rather than a growing prefix. Each claim is judged once rather than re-scored as the transcript accumulates. And abstention is invisible: its over-inference rate is a ratio over *emitted* claims, and no metric credits coverage. GISTBench [Fostiropoulos et al., 2026] is the nearest in stance: it verifies each inferred interest against the engagement signals cited for it rather than against a label, but has no abstention channel and scores a single snapshot per user. The neighbouring benchmarks fall on one side or the other of the same line. Some *reward* the unevidenced generalization we penalize. Some drop unevidenced fields from the question set, so honest abstention cannot be expressed. And some score an explicit `unknown` below a wrong guess. Appendix G works through each. Our stance descends from fact verification against retrieved evidence [Thorne et al., 2018] and from calibration work asking whether a model knows what it knows [Kadavath et al., 2022].

3 The Benchmark

Domain and persona. The domain is venue and event recommendation: restaurants, bars, cafés, and things to go to. Each subject is defined by a *persona card* of 24 fields in 8 domain groups, from hard constraints (dietary restrictions, allergies, accessibility, alcohol-free requirements) to taste, ambiance, logistics, and events (Appendix A). The groups match how a recommender uses them: hard constraints are filters (a venue is removed, not moved down the list); taste, ambiance, and quality signals are ranking features; and the three distribution fields (price band, party size, visit time) are what a booking step must fill in to hold a table at all. Each preference field is scored by an agreement score $a_f \in [0, 1]$ suited to its type: set overlap for the set-valued fields, distance along the scale for the ordinal fields, and a divergence for the three distributions over bins. The fidelity score (§6) averages a_f over the evidenced preference fields; the four hard constraints are checked separately, item by item.

Conversations. A subject's history is a sequence of dated sessions between the user and a recommendation-and-booking assistant: asking for somewhere to eat, reacting to suggestions, booking a table, reporting back afterwards. Sessions are written so that each field becomes knowable only where the schedule places evidence for it, and *fenced* so that no field leaks into a session that was never meant to reveal it; some sessions are distractors that reveal nothing new. Subjects run from 3 to 30 sessions (median 8), each about eight messages and 230 words. Section 4 describes the pipeline that produces them.

Serving protocol. Conversations are evaluated chronologically across discrete checkpoints, written $c_0, c_1, \ldots$ and counted from 0; in this corpus each checkpoint adds one session. At checkpoint t the system is given *only* the sessions new since $t - 1$ and must return the entire persona card, using an explicit `unknown` for any field it will not commit to; on a preference list field an empty list is a

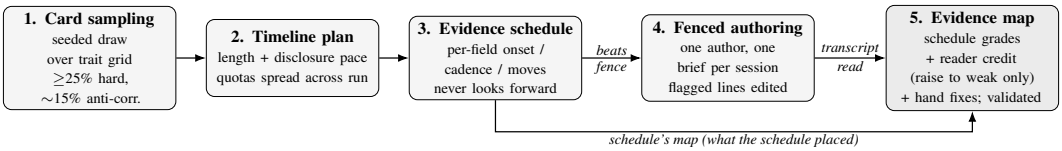


Figure 1: The corpus-construction pipeline. Stages 1–4 produce a subject's gold card and dated conversation; Stage 5 builds the evidence map from the grades the schedule placed, plus a small reader-credit layer and a few recorded hand corrections (§5). The transcript is served to the system and the held-out map to the scorer.

commitment ("none"), not an abstention. Two rules are built into the harness rather than left to the system:

1. *Earlier sessions are never re-served.* Carrying earlier text forward is the system's own choice. Replaying the full transcript is a different, easier benchmark.
2. *The answer key is never served.* The answer key is the gold card plus the evidence map (§5). The system sees transcripts and dates only; the schedule and the answer key stay held out.

A system under test is encapsulated behind a minimal stateful interface implementing `reset` and `observe(new_sessions)` $\rightarrow$ `card`. Any memory design can implement it.

4 Corpus Construction

Every subject is built by the five-stage pipeline of Figure 1. The plan (Stages 1–3: card, timeline, and evidence schedule) is fixed by a seed, the session text written from it in Stage 4 is released as a frozen copy with checksums, and the answer key of Stage 5 starts from the plan (§5).

Stage 1: Card sampling. A persona card is *drawn* at random, from a seed, over a declared grid of trait values, not asked of a model. Drawing rather than prompting lets us *build in* coverage instead of checking it afterwards. The draw gives a spread across the grid that can be audited, and meets two quotas. At least 25% of subjects are hard cases of one of four kinds: a stated preference that the scheduled behaviour contradicts, a hard constraint guaranteed to come up only once, with the session text checked so it is never raised again, a two-peaked price or visit-time distribution, or a few strong traits set against mid-scale values everywhere else. About 15% are anti-correlated cases: they intentionally violate one of the declared correlation clusters (C1 exploration, C2 spend, C3 sociality, C4 deliberateness; details in Appendix A) by a measurable margin, so a system cannot score by assuming clusters always hold together. No model is involved in this stage: a fixed rule check rejects a card whose values contradict each other or that gives the author nothing distinctive to voice, and never rejects one for being atypical.

Stage 2: Timeline planning. Each subject is assigned a conversation length and a pace at which it reveals itself. There are no difficulty tiers: evidence is thin early in every timeline and dense late, so a system passes through every density of evidence within a single subject.

Stage 3: Evidence schedule. This stage decides *where* each field becomes knowable. Every field has its own plan with three parts: an onset (the session it first surfaces in, drawn from an exponential distribution whose rate is the subject's pace), how often it recurs, and a sequence of moves that go from narrowing (a reaction, an anecdote) to pinning the value down. The one rule the scheduler enforces is that **it never looks forward**: the *beats* of session i (the pieces of evidence planned for it) depend only on the seed, the subject id, and the sessions before i, never on the total length. So a timeline can be extended without changing what came before: raising a subject from 8 to 25 sessions leaves the first 8 unchanged. It is also where the UNEVIDENCED band comes from: a field with a late onset simply never surfaces in a short timeline.

Stage 4: Fenced authoring. One author model (Claude Opus 5 here) turns the schedule into dated sessions. Each session is written from its own brief (its beats, its date, and the date of the previous conversation), never from the text of earlier sessions, so what a session is meant to reveal is set by the schedule alone. The corpus contains no scripted changes of mind, so a value once evidenced stays true and a grade never has to be walked back. The author is shown the whole person but *fenced*: it may reveal only the beats the schedule placed in that session. The fence keeps the unevidenced

band honest. Without it, an in-character aside ("I'll have the extra chilli") would leak a field (spice tolerance) the key records as unknown. After authoring, two LLM *fence checkers* checked every session, and an editor model (Claude Opus 5) rewrote only the messages they flagged as breaking the fence or the brief, leaving the rest of the session untouched; a small number of sessions were fixed by hand. 293 of the 612 released sessions carry at least one such edit.

Stage 5: The evidence map. The grades the schedule placed are the starting evidence map. Two LLM *evidence readers* on different models (Claude Opus 5 and Sonnet 5 here) then read every session for evidence the schedule did not place; §5 gives the rules for what they may change.

5 The Evidence Map

The evidence map is what makes calibration checkable. Each cell grades one field at one checkpoint by what the sessions served so far show: STRONG if they pin the value, WEAK if they bear on it without pinning it, and UNEVIDENCED if nothing in them bears on it yet. The map's core, the *schedule's map*, is computed from the schedule. Any evidence makes a field at least WEAK; it becomes STRONG once enough pinning evidence has accrued. One direct request, or a rejection that says what was wrong, pins an ordinal field or a hard constraint; a set field needs two, and a distribution five occasions. A reaction to a visit or a passing anecdote narrows a field but never pins it. Evidence comes from three sources:

- a **beat**, the field's own evidence in the subject's voice, which can raise a grade and is weighted by how strong it is;
- an **occasion**, the party size, time, and price band a session must state simply to book a table, which counts as incidental evidence for the matching distribution field whether or not a beat asked for it;
- a **trait implied by an occasion**, credited only up to WEAK and only where the field has no evidence of its own (e.g. a party of one is weak evidence for solo-dining comfort). The credit says only that the field has been narrowed, not which way, since an anti-correlated subject can break the usual pairing.

The schedule's map has two properties by construction: it is **prefix-stable** (appending future sessions cannot alter past grades, because the scheduler never looks forward) and **monotone** (it only adds evidence, so a field's grade never decreases across checkpoints). Figure 3 (Appendix A) shows the map for one 16-session subject: bands only ever darken left to right, six fields the schedule never touches stay unevidenced for the whole timeline, and each cell fixes whether committing or abstaining is the right move at that checkpoint. Appendix A traces one session through these rules.

What counts as evidence. The key, the reference prompts, and every LLM reader apply one written policy (stated in full in the prompts of Appendix E). Only what the user says, orders, chooses, or rejects counts; what the assistant offers or describes does not, even about a place the user then books. Booking a venue is not liking its cuisine, and a companion's needs are not the user's. Where a claim and the user's behaviour disagree, the behaviour counts. A value is credited under the word the user used, not a neighbouring one (sushi is not `japanese`).

Reader credit and hand corrections. Two small, recorded layers sit on top of the schedule's map. A page can carry more than its beats (booking a table for tonight states a planning horizon), so a cell the schedule left UNEVIDENCED is raised to WEAK where both evidence readers find the field readable on the page and the gold card holds a value for it; where only one finds it, or either flags a leak or a contradiction, a human rules on the full transcript. An evidence reader can never lower a grade, raise one to STRONG, or introduce a value the card does not hold. Separately, cells found to be keyed wrongly were corrected by hand against the transcript, each as a recorded edit to one field from a stated checkpoint on, and 86 gold-card values in 45 subjects were corrected by hand the same way, most of them by removing a list item the transcript never voiced. Every STRONG cell is still the schedule's own. Of the 14,688 field-checkpoint cells, 4,464 are STRONG, 5,244 WEAK, and 4,980 UNEVIDENCED; 225 (1.5%) differ from the schedule's grade (breakdown in Appendix A). Every score in this paper is on this final key.

Validation. A finished map must pass a consistency check before it can serve as an answer key: a field graded knowable at checkpoint t must remain at least that knowable at $t + 1$, so a system is never asked to know a fact and then to have forgotten it. A reader-credit raise comes only from a session already served, so prefix-stability is kept. All 50 released subjects pass.

6 Scoring: Fidelity, AUFC, and Skill

Fidelity covers the 20 preference fields. The four hard-constraint fields never enter it: they are scored separately by the hard-constraint failure rate (below), so a constraint error is not charged twice. Let $S(t), W(t), U(t)$ be the sets of preference fields graded strong, weak, and unevidenced at checkpoint t, with one exclusion: a distribution field graded weak is left out of all three, because a histogram's shape cannot be read from the handful of occasions behind a weak grade, so it is neither scored nor counted as a fabrication. For a field f the system under test either commits a value, scored by its metric $a_f \in [0, 1]$ against the gold card in force at checkpoint t, or emits `unknown`. The gold at t is not always the final card: for the four cuisine and event-type lists it holds only the items the conversation has shown so far, and for five ordinal fields it is the highest level the user's turns support so far (fields listed in Appendix A); every other field is scored against the final card. The per-band rule is:

Band	Commit a value	Emit `unknown`
$S(t)$ strong	scored a_f	scored 0 (a miss)
$W(t)$ weak	scored a_f	scored WEAK-CREDIT $= 5/6$
$U(t)$ unevidenced	$-\lambda$ penalty (fabrication)	scored 0 (correct abstention)

The constant WEAK-CREDIT $= 5/6$ is set so that a system reading a weak field exactly as far as the evidence goes gains nothing by either committing or abstaining. It is derived for four-level ordinal fields, 13 of the 20 scored: when the evidence narrows such a field to two adjacent levels, a 50/50 guess between them scores $\frac{1}{2}(1 + \frac{2}{3}) = 5/6$ on average, so committing pays only if the system has signal beyond the narrowing. The set fields take the same constant by convention; Appendix C varies it by ± 0.08. Let $\mathrm{score}(f, t)$ denote the per-field score for evidenced fields $f \in S(t) \cup W(t)$ in the table, and let $C_U(t) = \{f \in U(t) : f \text{ is committed}\}$ be the set of fabrications. The fabrication penalty is

$$\mathrm{Pen}(t) = \lambda\, |C_U(t)|, \qquad \lambda = 1.$$

With $\lambda = 1$ a fabrication costs exactly what a miss on a strong field costs, so one fabrication cancels one correct commit. The value is a stated position, not a fitted one (§8); every gap reported in §7 is rechecked under $\lambda \in \{0.5, 1, 2\}$, crossed with the other two scorer settings, and those reported as real hold under all 27 (Appendix C). The fidelity at checkpoint t is therefore

$$F(t) = \frac{\sum_{f \in S(t) \cup W(t)} \mathrm{score}(f, t) - \mathrm{Pen}(t)}{|S(t)| + |W(t)|}.$$

Its denominator is fixed by the map rather than by the system's decisions. Let $T_e = \{t : |S(t)| + |W(t)| \geq 7\}$ be the eligible checkpoints (5 and 10 are checked in Appendix C), so the headline number is not dominated by thin early checkpoints. The **Area Under the Fidelity Curve (AUFC)** is

$$\mathrm{AUFC} = \frac{1}{|T_e|} \sum_{t \in T_e} \min\big(1, \max(0, F(t))\big).$$

Skill is normalized per checkpoint. Because abstaining on a weak field earns credit, a card that abstains everywhere scores not 0 but $F_0(t) = \text{WEAK-CREDIT} \cdot |W(t)|/(|S(t)| + |W(t)|)$, and skill rescales the unclipped $F(t)$ against that floor:

$$\mathrm{skill}(t) = \frac{F(t) - F_0(t)}{1 - F_0(t)}.$$

A system that always emits `unknown` therefore scores exactly 0 at every checkpoint, and an *oracle* that commits the gold value wherever the map grades a field evidenced scores 1, so negative skill has a concrete meaning: *worse than emitting nothing*. A subject's **skill** is the mean of $\mathrm{skill}(t)$ over its eligible checkpoints, and the reported skill is the mean over subjects with at least one eligible checkpoint. Skill is therefore not AUFC rescaled after averaging, and, being unclipped, it can fall well below -1. We report skill; the paired significance tests compare per-subject AUFC (Appendix C).

Hard constraints. The **hard-constraint failure rate** is the share of subjects with at least one checkpoint on which the emitted card misses or invents a member of one of the four hard-constraint

fields. A missed member counts only once the field is evidenced; an invented one counts at any checkpoint. An `unknown` on an evidenced constraint field with a non-empty gold value is a miss: a filter failure in its own right, reported beside skill and never averaged into fidelity.

Zero-cost baselines as checks. Four baselines that need no model calls check the key and harness: the oracle and `abstain` land exactly on their anchors, and the population prior and random land near skill -4, fabricating on every unevidenced field (Appendix B).

7 Experiments

What these numbers are for. The measurements below are reference points, not a leaderboard: they show that the score separates real systems, that its anchors sit where the construction says they do, and that the failures it penalizes are ones systems like those deployed today actually make (caveats in §8). Each *(system, backbone)* cell is a single run; the paired confidence intervals (Appendix C) resample *subjects* within it, so they cover variation across subjects, not run-to-run randomness.

Systems under test. We test three deliberately simple reference systems that differ in call budget and carried state (diagrams in Appendix D). `prompted` spends one call per checkpoint: it is shown the card it last emitted plus the new sessions and emits the card again, so the card is all the model carries forward. `notebook` is the same loop with a free-text scratchpad it writes to itself and is shown back, testing whether free-form notes buy anything over the card alone. `memory` spends two calls: an *extract* call reads only the new sessions and writes candidate memories, and an *integrate* call, which never sees the conversation, merges them into a persistent bank and rebuilds the card from it. All three carry state across checkpoints. We call `prompted` and `notebook` *single-pass* because one call reads the new sessions and writes the card; we call `memory` *stateful* because its card is rebuilt from a separate bank by a call that never sees the conversation. All three also carry a running count of occasions for the three distribution fields, held by code rather than by the model (Appendix D). Because earlier sessions are never re-served (§3), what each system carries beyond those counts is exactly what it chooses to hold. All three run on fixed system prompts that state the evidence policy of §5; the field menu comes in the user turn of every call except `memory`'s extract call (Appendix E).

The two single-pass systems are run on four frontier backbones: GPT-5.6 Luna, Gemini 3.7 Flash, DeepSeek V4 Pro, and GLM-5.3 Flash; `memory` is run on GPT-5.6 Luna only (routes, providers, and reasoning settings in Appendix D). A results row is a *(system, backbone)* pair, because the same design behaves differently on different backbones.

Reference measurements. Figure 2 reports skill by system and backbone over all 50 subjects (47 enter the mean; three never reach the seven evidenced fields an eligible checkpoint needs, §6); Appendix C gives the same numbers as a table. **The single-pass extractors are well calibrated**: every bar clears the abstain floor by a wide margin, from a little under half the distance to the oracle (GLM-5.3 Flash) to about four fifths (Gemini 3.7 Flash). **The backbone matters more than the system**: changing backbone moves skill by up to 0.36, changing between the two single-pass systems on the same backbone by at most 0.06. **A free-text scratchpad provides no consistent benefit**: the `prompted`/`notebook` order flips by backbone, and only the Gemini 3.7 Flash gap survives a paired bootstrap over subjects on AUFC and holds under every scorer setting we tried (Appendix C). A free-text scratchpad therefore neither reliably helps nor hurts a single-pass system.

For the stateful system we have a single cell, reported in text rather than in Figure 2: `memory` on GPT-5.6 Luna scores $+0.493$, against $+0.669$/$+0.700$ for the single-pass systems on the same backbone, with a hard-constraint failure rate of 0.30 against 0.08–0.14 (`abstain`: 0.56). The gap to both single-pass systems is significant and holds under every scorer setting (Appendix C). It still clears the abstain floor, so on this backbone the memory layer costs skill but stays above silence; §7.1 says what it loses instead.

7.1 What the systems get wrong

The score says *how much* a system fabricates; the released transcripts let us ask *what kind* of mistake each one is. We grouped every cell the scorer charged (a commit scoring below 0.5, a fabrication, or a missed or invented constraint item; an `unknown` on a strong preference field is not included) on the nine runs of §7 (each on all 50 subjects) into 807 *episodes*; a field wrong with the same value across consecutive checkpoints counts once. LLM *failure readers*, blind to which system produced

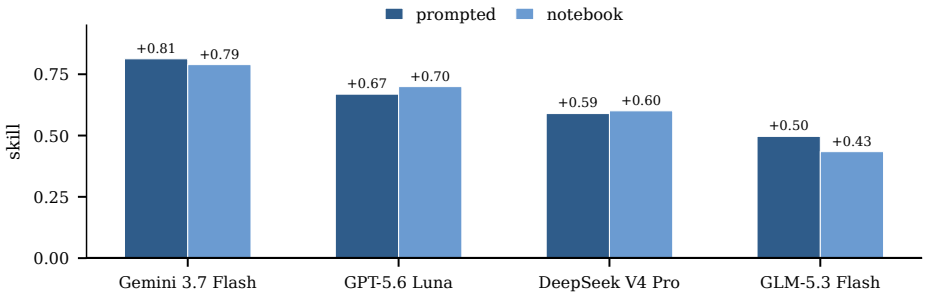


Figure 2: Skill (`abstain` = 0, `oracle` = 1) of the two single-pass systems by backbone; 50 subjects. The stateful system (`memory`) is reported in the text.

an episode, read each distinct case (episodes sharing subject, field, values, and first checkpoint are read once) against the full served transcript under the answer key's written evidence policy, and every case not judged a genuine error was read a second time, independently (Appendix F). The *confirmed mistakes* (episodes judged genuine system errors rather than artifacts of the key) show five patterns.

Committing too early is the main failure, and it does not pay. Between 49% (DeepSeek V4 Pro) and 74% (GLM-5.3 Flash) of each backbone's single-pass confirmed mistakes are commits on a field the transcript has not yet evidenced, 45% of them at the very first checkpoint. The guess is in the final gold card in at most 17% of episodes, so even checked against the hidden card, the guess is mostly wrong.

The guess is the middle of the scale. On the ordinal fields, 266 of 323 errors are commits on an unevidenced field, and the level committed is the middle of the scale whatever the persona (level codes in Appendix A): `notices_value` for *deal-seeking*, `fine_with_lively` for *noise*, one of the two middle levels for *design*. One mechanism does lean one way: every backbone reads *formality ceiling* off the *current* occasion, so one counter-service request at c_0 becomes the ceiling.

Which cells fail varies by backbone; what triggers a failure is shared. Of the (subject, field) cells that any single-pass backbone fails on `prompted`, 64% fail under one backbone only and 5% under all four. What is shared is the *trigger*: a per-head budget cap becomes a deal-seeking level; a format chosen or praised becomes an event type (`market`, the most-invented event type, on all nine runs); a soft preference becomes a hard constraint ("a proper table rather than perching" $\rightarrow$ `seating_required`, 74 episodes). Nearly every invented list element we traced has a line in the text behind it before the commit; the failure is in deciding what counts as evidence, not in remembering. Some of these lines are judgment calls a human could draw differently (a proper table rather than perching read as a seating need); the key settles each by one written policy, which the prompts state.

A single-pass system can also throw away what it holds. Told to record only what the person says, some backbones apply that to the *carried card* as well as the new batch. At such a *wipe*, five or more held fields are rewritten to `unknown` at once and the card is rebuilt from the current sessions, though nothing in them contradicts what was held. `prompted` wipes 16 times on DeepSeek V4 Pro and never on the other three backbones; `notebook` wipes 3–6 times on every backbone. Hard constraints suffer most because most are mentioned only once: 33 of DeepSeek V4 Pro `prompted`'s 42 constraint drops sit inside a wipe. A prompt for such a system should say that what it holds stands unless contradicted.

A memory layer changes the kind of failure, not only the amount. The one stateful run (`memory` on GPT-5.6 Luna) has more than 3$\times$ the confirmed mistakes of the single-pass systems on the same backbone (3.14 per subject against 0.84–0.90), and its mix shifts toward loss: 36% of its episodes are losses (a held constraint gone in a session that does not mention it, a distribution frozen on one occasion, a repeatedly evidenced value overwritten by one contrary line), against 11–14% for the single-pass systems on that backbone. Single-pass errors persist (median length 3–5 checkpoints; the wrong value rides forward on the carried card) while stateful errors flicker (median length 1; 20% of its wrong cells go wrong, right, wrong again, against 0–3% single-pass on the same backbone). The raw outputs of the two calls point to the protocol rather than the backbone, though one backbone cannot confirm it: the card is re-read from the bank at every checkpoint with no previous card shown, so a vague memory ("enough room to turn") is mapped onto the schema afresh each time;

contradictions are *added* to the bank rather than superseding what they contradict; and 42% of the distribution data points match no bin and are dropped, starving those fields of data. That last one is a gap in our reference implementation, not in memory layers as such: the extract call is never shown the bin names (Appendix E). The same unstable re-read, not a single early drop, is behind its higher hard-constraint failure rate (§7).

8 Limitations and Future Work

We score the profile, not the recommendations. DossierBench measures the fidelity of a preference profile against what the conversation evidenced; it produces no ranked list of venues and evaluates no recommendation. The assumption that a better-calibrated profile yields better recommendations, and in particular that a fabricated field costs more downstream than an abstention, motivates the score but is not measured here. The fabrication penalty should be read as a stated position on what a profile owes a recommender, not as a measured estimate of the harm done downstream.

Coverage of the measurements. The single-pass systems are measured on all 50 subjects and all four backbones; the stateful layer is measured on one backbone (`memory` on GPT-5.6 Luna), and each cell is a single run. The failure-mode reading of §7.1 relies on LLM failure readers (Appendix F). They found the answer key itself wrong on six fields: four are corrected in the released key, and two remain (2 of the 807 episodes, 0.2%; Appendix F). They also doubted 55 more gold questions (single-reader calls, kept as keyed; Appendix F); a human review of the key is next. One credit rule also reads the hidden card: an outing counts as evidence for the formality ceiling only when it is at the subject's ceiling (about 240 cells); a rule that reads the page alone is planned for the next release. All four backbones are general-purpose frontier models and the reference systems are simple; a purpose-built memory system (for example, one whose extract call is shown the field menu) could close or reverse the gap. `memory` also differs from the single-pass systems in several ways at once (Appendix D), so the gap cannot be laid on any one of them. The reference systems emit the 24-field card directly; a profile held as prose or embeddings needs an adapter we do not yet provide. The reference prompts state the answer key's evidence policy (Appendix E), so scores partly measure how well a system follows a stated policy, not only how well it judges evidence unaided. We disclose the policy on purpose, so that a calibration failure is not a failure to guess the grader's rules; how much a prompt without the policy loses is untested.

Scope of the corpus. The corpus is synthetic and deliberately limited to one recommendation domain, venues and events, because that is what the 24-field schema is designed around. Whether the benchmark carries over to domains whose preference fields look different (media, retail, travel) is untested; real user histories are noisier and mix domains within a conversation; preferences are static, with no scripted change of mind; and whether the single-pass advantage survives conversations longer than our timelines is open. The three distribution fields are scored by a divergence that treats the bins as unordered, so it does not see how far a misplaced share has moved.

Future work. Five extensions address these limits: **preferences that change over time**, letting a grade go down so a documented change of mind is scored; **preferences that depend on the occasion**, graded per occasion rather than per field; **downstream impact**, costing each error by what a recommender then recommends; **ablations and repeated runs**: one design choice at a time, including prompt wording and the evidence policy, over several runs per cell; and a **schema-agnostic interface** that queries one field at a time, so a profile held as prose or embeddings can be scored.

9 Conclusion

The preference profile behind a recommender should be judged on whether it knows when it doesn't know: a fabricated field is a standing bias on every later suggestion, while an abstention is something the recommender can act on. DossierBench makes that checkable by pairing every synthetic subject with a schedule-derived, consistency-checked evidence map and scoring commitment against it. On 50 subjects and four backbones, single-pass extractors reach skill 0.43–0.81, and the backbone moves that number more than five times as far as the choice between them; the one stateful layer we scored trails both at twice the call budget; and the dominant failure is committing before the evidence arrives. We release the corpus, scorer, and reference systems, and hope the evidence map is reused wherever "correct abstention" needs to be a checkable claim.

Acknowledgments and Disclosure of Funding

This work was carried out at Soko and received no specific funding. The authors declare no competing interests.

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A The 24 persona-card fields

Table 1 lists every field of the persona card with the group it belongs to in the text of §3, its type, its agreement score a_f, and its allowed values. Every field also accepts the literal `unknown`, the abstention the scoring rule of §6 acts on. The four hard-constraint fields are *filters*: a recommender uses them to rule venues out, and `allergies` is safety-critical. The *Corr.* column gives the *correlation cluster* (C1 exploration, C2 spend, C3 sociality, C4 deliberateness) that the card sampler of §4 uses to draw consistent subjects and, for the anti-correlation quota, to deliberately break; fields outside any cluster are drawn independently. Figure 3 illustrates how these 24 fields evolve across checkpoints for an example subject.

Metrics. For a set field, $a_f = |P \cap G|/|P \cup G|$ with two empty sets scoring 1. For an ordinal field with four levels, $a_f = 1 - d/3$ where d is how many steps apart the two levels are, so a miss by one level scores $2/3$ and a miss by the whole scale scores 0. For a distribution over bins, $a_f = 1 - \mathrm{JSD}_2(P \,\|\, G)$ with the Jensen–Shannon divergence taken in base 2 so that it lies in $[0, 1]$. A hard-constraint field has no a_f: it is checked item by item (§6).

Gold at a checkpoint. For `cuisines_liked`, `cuisines_avoided`, `event_types_liked`, and `event_types_avoided`, the gold at checkpoint t holds only the items the conversation has shown so far. For `solo_dining_comfort`, `formality_ceiling`, `deal_seeking`, `novelty_vs_loyalty`, and `noise_preference`, it is the highest level the user's turns support so far. Every other field is scored against the final card.

Table 1: All 24 fields of the persona card. *Corr.* is the correlation cluster used by the card sampler; *Values* summarizes the allowed values (ordinal level codes with their wordings, and set vocabularies, are given in full below).

Group	Field	Type	a_f	Corr.	Values (ordinals: lowest . . . highest level)
Hard constraints	`dietary`	set	item check	—	subset of 9-item vocabulary
	`allergies`	set	item check	—	subset of 9-item vocabulary
	`accessibility_needs`	set	item check	—	subset of 9-item vocabulary
	`alcohol_free_required`	boolean	item check	—	true / false
Taste	`cuisines_liked`	set	Jaccard	C1	subset of 39-item vocabulary
	`cuisines_avoided`	set	Jaccard	—	subset of 39-item vocabulary
	`spice_tolerance`	ordinal	$1 - d/3$	—	avoids_heat . . . seeks_hottest
	`adventurousness`	ordinal	$1 - d/3$	C1	sticks_to_known . . . seeks_unidentifiable
Economics	`price_band_distribution`	distribution	$1 - \mathrm{JSD}_2$	C2	4 bins
	`deal_seeking`	ordinal	$1 - d/3$	C2	ignores_price . . . plans_around_specials
Ambiance	`noise_preference`	ordinal	$1 - d/3$	C3	needs_quiet . . . wants_loud
	`formality_ceiling`	ordinal	$1 - d/3$	C2	counter_service . . . white_tablecloth
	`design_sensitivity`	ordinal	$1 - d/3$	C2	doesnt_notice . . . room_over_food
Social	`party_size_distribution`	distribution	$1 - \mathrm{JSD}_2$	C3	4 bins
	`solo_dining_comfort`	ordinal	$1 - d/3$	C3	wont_eat_alone . . . enjoys_solo
Logistics	`travel_radius`	ordinal	$1 - d/3$	—	walking_distance . . . out_of_town
	`visit_time_distribution`	distribution	$1 - \mathrm{JSD}_2$	—	5 bins
	`planning_style`	ordinal	$1 - d/3$	C4	fully_spontaneous . . . books_weeks_ahead
	`wait_tolerance`	ordinal	$1 - d/3$	C4	leaves_if_line . . . queue_is_part_of_it
Events	`event_types_liked`	set	Jaccard	—	subset of 18-item vocabulary
	`event_types_avoided`	set	Jaccard	—	subset of 18-item vocabulary
	`participation_vs_spectating`	ordinal	$1 - d/3$	—	watches_only . . . seeks_hands_on
Quality signals	`rating_reliance`	ordinal	$1 - d/3$	C4	ignores_ratings . . . threshold_gated
	`novelty_vs_loyalty`	ordinal	$1 - d/3$	C1	same_few_places . . . almost_never_repeats

Ordinal levels. Each ordinal field has four discrete levels ($0 \le \ell \le 3$) defined by behaviour, listed here from lowest (**0**) to highest (**3**). Each level is given with the code a system emits followed by its gloss in the prompts, so that a level is pinned by behaviour in the transcript rather than by an adjective.

`spice_tolerance` **0**: `avoids_heat` (*avoids heat entirely*) → **1**: `mild_only` (*mild only; will ask for it toned down*) → **2**: `likes_real_spice` (*likes real spice and orders for it*) → **3**: `seeks_hottest` (*seeks the hottest thing on the menu*).

`adventurousness` **0**: `sticks_to_known` (*sticks to dishes they already know*) → **1**: `familiar_variations` (*tries variations of the familiar*) → **2**: `orders_unfamiliar` (*orders the dish they don't recognize*

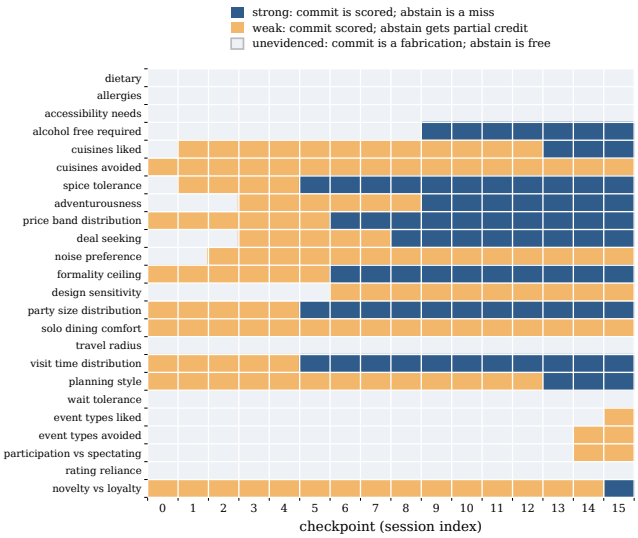


Figure 3: The evidence map for one subject (24 fields × 16 checkpoints, c_0–c_{15}). No reader credit or hand correction changes a grade on this subject, so its bands are exactly the schedule's map (§5); each cell is what a careful reader could know so far. Grades never go down, so bands only darken rightward, and the six fields the schedule never evidences stay light throughout.

while the table plays safe) → **3**: `seeks_unidentifiable` (*picks the place* because *they can't read the menu*).

`deal_seeking` **0**: `ignores_price` (*ignores price*) → **1**: `notices_value` (*notices value but doesn't act on it*) → **2**: `looks_for_deals` (*checks for a discount once they have already chosen the place*) → **3**: `plans_around_specials` (*lets the happy-hour window pick the place and the hour*).

`noise_preference` **0**: `needs_quiet` (*needs it quiet enough to talk*) → **1**: `prefers_low_hum` (*prefers a low hum*) → **2**: `fine_with_lively` (*fine with lively*) → **3**: `wants_loud` (*wants it loud and busy*).

`formality_ceiling` **0**: `counter_service` (*counter service only*) → **1**: `casual_sit_down` (*casual sit-down*) → **2**: `smart_casual` (*smart-casual*) → **3**: `white_tablecloth` (*white tablecloth*).

`design_sensitivity` **0**: `doesnt_notice` (*doesn't notice the room*) → **1**: `mild_preference` (*mild preference for a nice room*) → **2**: `room_matters` (*rules out a place whose room they dislike when there is an alternative*) → **3**: `room_over_food` (*will pick a worse meal for a better room*).

`solo_dining_comfort` **0**: `wont_eat_alone` (*won't eat alone*) → **1**: `takeaway_or_counter` (*alone only for takeaway or at a counter*) → **2**: `comfortable_at_table` (*comfortable at a table alone*) → **3**: `enjoys_solo` (*actively enjoys solo meals*).

`travel_radius` **0**: `walking_distance` (*only what they can walk to in ten minutes*) → **1**: `own_neighborhood` (*one short bus or subway ride, never across town*) → **2**: `anywhere_in_city` (*will cross the city for it, but not leave it*) → **3**: `out_of_town` (*will get on a train out of town for one meal*).

`planning_style` **0**: `fully_spontaneous` (*decides on the pavement, already out*) → **1**: `decides_same_day` (*picks in the morning for that evening*) → **2**: `books_days_out` (*books a few days out*) → **3**: `books_weeks_ahead` (*books weeks ahead*).

`wait_tolerance` **0**: `leaves_if_line` (*turns around if anyone is waiting ahead of them*) → **1**: `fifteen_min_max` (*waits a quarter of an hour, then leaves*) → **2**: `hour_for_the_right_place` (*will stand in an hour-long line once they are already there*) → **3**: `queue_is_part_of_it` (*plans the evening around being early in the queue; a wait is never a reason to go elsewhere*).

`participation_vs_spectating` **0**: `watches_only` (*watches only*) → **1**: `joins_if_pushed` (*takes part only when asked directly*) → **2**: `joins_readily` (*joins in unasked at whatever event they happen to be at*) → **3**: `seeks_hands_on` (*won't book an event where they would only be watching*).

`rating_reliance` **0**: `ignores_ratings` (*ignores ratings*) → **1**: `glances_at_them` (*glances at them*) → **2**: `checks_before_going` (*checks ratings before going*) → **3**: `threshold_gated` (*won't go below a rating threshold*).

`novelty_vs_loyalty` **0**: `same_few_places` (*returns to the same few places*) → **1**: `mostly_regulars` (*mostly regulars, occasional new*) → **2**: `mostly_new` (*mostly new places, some regulars*) → **3**: `almost_never_repeats` (*almost never repeats a venue*).

Closed vocabularies for set fields. A set field's value is any subset of its vocabulary (the empty set included), and a system may only commit items from the vocabulary.

`dietary` `vegetarian`, `vegan`, `pescatarian`, `halal`, `kosher`, `gluten_free`, `dairy_free`, `low_sodium`, `keto`.

`allergies` `peanuts`, `tree_nuts`, `shellfish`, `fish`, `eggs`, `milk`, `soy`, `wheat`, `sesame`.

`accessibility_needs` `step_free_access`, `accessible_restroom`, `wide_aisles`, `low_lighting_ok`, `quiet_room_needed`, `service_animal`, `seating_required`, `braille_menu`, `assistive_listening`.

`cuisines_liked`, `cuisines_avoided` (39 items, grouped by region/type):

- *East & SE Asian:* `chinese`, `dim_sum`, `filipino`, `japanese`, `korean`, `malaysian`, `ramen`, `sushi`, `taiwanese`, `thai`, `vietnamese`.
- *European & Mediterranean:* `french`, `german`, `greek`, `italian`, `mediterranean`, `moroccan`, `portuguese`, `spanish`, `turkish`.
- *Americas & Casual:* `american`, `bbq`, `burgers`, `cajun`, `caribbean`, `deli`, `latin_american`, `mexican`, `peruvian`, `soul_food`, `steakhouse`.
- *South Asian, Middle Eastern, African & Other:* `brunch`, `ethiopian`, `indian`, `lebanese`, `middle_eastern`, `pizza`, `seafood`, `vegan_restaurant`.

`event_types_liked`, `event_types_avoided` `art_opening`, `book_reading`, `comedy`, `cooking_class`, `dance_night`, `film_screening`, `food_festival`, `karaoke`, `live_jazz`, `live_rock`, `market`, `museum_late`, `open_mic`, `pub_quiz`, `supper_club`, `tasting_menu`, `theater`, `wine_tasting`.

Distribution bins. `price_band_distribution` is over per-head spend (<£15, £15–30, £30–50, £50+); `party_size_distribution` over party size (1, 2, 3–5, 6+); `visit_time_distribution` over occasion (weekday lunch, weekday evening, weekend day, weekend evening, late night). A distribution is kept, rather than the single most common bin, because it preserves two-peaked behaviour (cheap weeknights and occasional blowouts) that a single value would flatten.

Reader credit and hand corrections in numbers. The reader credit of §5 raises 181 cells in 28 subjects from UNEVIDENCED to WEAK (170 on both evidence readers, 11 on one and a human ruling) and lowers none. The hand corrections leave 44 cells in 9 subjects different from the schedule's grade (29 raised to WEAK, 15 lowered to UNEVIDENCED), and correct 86 gold-card values in 45 subjects, 67 of them by removing a list item the transcript never voiced.

A worked example. The first session of subject `gen-000` opens: "*Looking for somewhere to eat tonight, just me. Not planning on spending much — under fifteen quid ideally.* [...] *Somewhere I can just order at the counter* [...] *I just don't want to be crossing the whole city for it.*" Its one beat (the last line, a reaction) makes `travel_radius` WEAK. Its occasion (a party of one, under £15, a weekend evening) makes the three distributions WEAK (too few occasions to score yet, §6) and, as implied traits, `solo_dining_comfort`, `formality_ceiling` (a counter-service outing, credited because counter service is this subject's ceiling; an outing below a subject's ceiling credits nothing, so the same request is a fabrication in the subjects of §7.1) and `planning_style` (a same-day plan). The other 17 fields stay UNEVIDENCED at c_0, including `spice_tolerance`, whose gold value is `seeks_hottest`: committing it here is a fabrication even though it is right.

B The zero-cost baselines

Table 2 gives the four zero-cost baselines of §6 in full. None of the four reads a transcript, so their values are fixed by construction and are run before any model as a check on the key and harness. The `oracle` commits the gold value on every field the map grades STRONG or WEAK and abstains elsewhere; `abstain` emits `unknown` on every field; `population` commits the most common value in the corpus on every field at every checkpoint; `random` commits a value drawn uniformly at random. Skill is fidelity rescaled at each checkpoint so that `abstain` $= 0$ and `oracle` $= 1$, then averaged over eligible checkpoints (§6), so the first two rows are the anchors themselves.

The oracle's fidelity is exactly 1.000 at every checkpoint of every subject, so the gold card and the map agree. Both always-committing baselines fabricate on the same 3,908 cells, which is every UNEVIDENCED preference-field cell at any checkpoint, and at $\lambda = 1$ both land near skill -4. `abstain` has a hard-constraint failure rate of 0.56 because an `unknown` on an evidenced constraint field with a non-empty gold value counts as a miss (§6).

Table 2: The zero-cost baselines fix the scorer's ceiling and floor. All 50 subjects (612 sessions); 47 enter the AUFC and skill means (§6). All four values are fixed by construction. Both always-committing baselines lose to silence.

Baseline	mean AUFC	skill	fabrications	hard-constraint failure rate
`oracle` (ceiling)	**1.000**	1.000	0	0.00
`abstain` (floor)	0.629	**0.000**	0	0.56
`population`	0.117	−3.969	3,908	0.56
`random`	0.114	−3.943	3,908	1.00

C Skill by system and backbone

Table 3 gives the numbers plotted in Figure 2, plus the single stateful cell reported in the text of §7, over the same subjects.

Paired comparisons. Gaps are tested in AUFC, not in the skill the table reports, by a paired bootstrap over subjects; ΔAUFC is the first system's AUFC minus the second's, and the intervals are 95%. Each gap is also recomputed under all 27 combinations of $\lambda \in \{0.5, 1, 2\}$, WEAK-CREDIT $\in \{5/6 - 0.08, 5/6, 5/6 + 0.08\}$, and an eligibility threshold of 5, 7, or 10 evidenced fields.

- Gemini 3.7 Flash, `prompted` − `notebook`: $+0.009$, CI $[+0.001, +0.017]$; `prompted` is ahead under all 27 settings.
- GLM-5.3 Flash, `prompted` − `notebook`: $+0.015$, CI $[-0.005, +0.034]$; borderline.
- GPT-5.6 Luna and DeepSeek V4 Pro, `prompted` − `notebook`: the intervals straddle zero and the sign changes across settings.
- GPT-5.6 Luna, `memory` − `prompted`: -0.065, CI $[-0.084, -0.044]$; `memory` − `notebook`: -0.067, CI $[-0.082, -0.052]$; both hold under every setting.

Table 3: Skill (fidelity rescaled per checkpoint so `abstain` $= 0$, `oracle` $= 1$; §6) by system and backbone, all 50 subjects. Higher is better; negative is worse than emitting nothing. The single-pass systems clear the floor by a wide margin on every backbone. The stateful system (`memory`) is scored on GPT-5.6 Luna only; the other three cells are unrun.

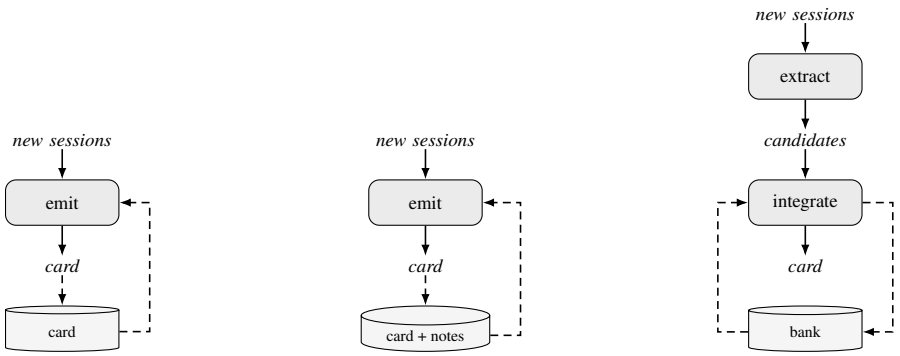
System	Gemini 3.7 Flash	GPT-5.6 Luna	DeepSeek V4 Pro	GLM-5.3 Flash
`prompted`	+0.813	+0.669	+0.590	+0.497
`notebook`	+0.789	+0.700	+0.601	+0.434
`memory`	—	+0.493	—	—

D The three systems under test

Figure 4 draws one checkpoint of each system of §7. Every system receives the same input at a checkpoint (the new sessions since the last checkpoint, never the earlier ones) and must emit a full persona card. What differs is the number of model calls spent and what is carried forward (the cylinders), which is the only thing the system sees of its own past at the next checkpoint. All three systems also carry a running count of occasions for the three distribution fields. The model reports one bin per occasion it sees, the code adds them up, and the card's distributions are the normalised counts, so a distribution is built from counted occasions rather than re-estimated from memory.

`prompted` (a) is one call. The model is shown the card it emitted at the previous checkpoint and the new sessions, and emits the card again. Beyond the occasion counts, its carried state is that card alone. Anything it wants to keep must therefore live in a card field; there is nowhere else to put it.

`notebook` (b) is the same loop with one addition: a free-text scratchpad the model writes at each checkpoint and is shown back at the next, alongside the last card. It uses the same call budget as `prompted` with more carried state, and tests whether free-form carried notes (reminders of weak evidence, context that does not fit a field) buy anything over the card alone.



(a) `prompted` (b) `notebook` (c) `memory`

Figure 4: The three systems under test, one checkpoint each (input *new sessions* on top, emitted *card* on the bottom). Boxes are model calls. Cylinders are carried state (dashed loops), the only thing a system sees at the next checkpoint since earlier sessions are never re-served. The single-pass pair (a, b) spend one call and carry the last card (`notebook` also its notes); `memory` (c) spends two, one to extract and one to integrate, and carries an explicit bank the card is re-read from.

`memory` (c) is two calls with a deliberate information barrier between them. The *extract* call reads only the new sessions and writes candidate memories; it never sees the bank, so it cannot rubber-stamp "nothing changed." The *integrate* call is shown the bank and the candidates, never the raw conversation. It decides for each candidate whether to add it, let it supersede an existing entry, or drop it as a duplicate, and then rebuilds the card from the updated bank. The carried state is the bank, plus the occasion counts, which grow from the extract call because it is the only one that reads the conversation. The handoff between the two calls loses information by design: whatever the extractor fails to write down is gone. That loss is the cost this system exists to measure. In our run it explains the starved distributions; most of the stateful failures of §7.1 come instead from the integrate call, which re-reads the card from the bank afresh at every checkpoint.

All three are reference implementations run on fixed system prompts, which are reproduced in Appendix E and ship with the benchmark.

Backbones and routing. All four backbones are called through OpenRouter at the fixed routes of Table 4. DeepSeek V4 Pro and GLM-5.3 Flash are pinned to one provider with fallbacks off, so every call hits the same endpoint; GPT-5.6 Luna and Gemini 3.7 Flash route by OpenRouter default. We send no reasoning-effort override, so the effort shown is each route's OpenRouter default, not a setting we chose. Every model in this paper, the four backbones and the Claude models that build the corpus and read the failures, is used through its provider's API under the provider's terms of use.

Table 4: Backbone routing. All calls go through OpenRouter. *Effort* is the route's OpenRouter default; *Reasoning* is whether chain-of-thought is on at that default, which we did not verify for DeepSeek V4 Pro.

Backbone	OpenRouter route	Provider	Effort	Reasoning
Gemini 3.7 Flash	`google/gemini-3.7-flash`	OpenRouter default	`medium`	always on
GPT-5.6 Luna	`openai/gpt-5.6-luna`	OpenRouter default	`medium`	on by default
DeepSeek V4 Pro	`deepseek/deepseek-v4-pro-0813`	Alibaba, no fallback	`high`	not verified
GLM-5.3 Flash	`z-ai/glm-5.3-flash`	Z.AI fp8, no fallback	`max`	always on

E System prompts

The three systems of Appendix D run on the system prompts below, reproduced verbatim from the released code. Each was written from the schema and the answer key's evidence policy. `prompted` and `notebook` share one system prompt (§E.1); `memory` has one per call (§E.2, §E.3). The per-checkpoint user turn differs by call. The single-pass calls get the 24-field menu with each value's gloss and bins (Appendix A), the last card (for `notebook` also its notes), the new sessions, an instruction to

report the three distribution fields as one data point per occasion seen in the new sessions rather than as estimated shares, and the JSON shape to emit. `memory`'s extract call gets only the new sessions, the same data-point instruction, and the JSON shape; it is never shown the field menu (§7.1). The integrate call gets the field menu, the bank, and the new candidates, and leaves the distribution fields to the counted data points.

E.1 `prompted` and `notebook`

```
You are building a preference profile of one person from their conversations with a restaurant-
recommendation assistant. You will be shown their conversations a few at a time, in order, and after each
batch you output the profile as it stands.

You are scored on CALIBRATION, not on coverage. For each field:

  - the conversations so far PIN the value  -> give the value. Saying "unknown" scores zero.
  - the conversations so far NARROW it      -> give your best value. "unknown" gets partial credit.
  - the conversations have NOT TOUCHED it   -> say "unknown". A value here is a FABRICATION and is
penalised EVEN IF IT HAPPENS TO BE CORRECT.

That last line is the whole task. Guessing from a stereotype, from a population base rate, or from another
 field ("they like Sichuan, so they must want it hot") earns nothing and costs you whenever the
conversations have not shown it. This dataset deliberately contains people who violate the usual
correlations, so the guess is not even reliably right.

Seven fields hold LISTS: dietary, allergies, accessibility_needs, cuisines_liked, cuisines_avoided,
event_types_liked, event_types_avoided. On these, an empty list [] is NOT the same as "unknown": it is a
positive claim that the person has NONE, scored as a commitment, so on a field the conversations have not
established it is a FABRICATION. Two rules:

  - dietary, allergies, accessibility_needs: write [] ONLY once the person has said they have none ("no
allergies", "no dietary needs"). Until they do, say "unknown".
  - cuisines_liked, cuisines_avoided, event_types_liked, event_types_avoided: never write [] -- say "
unknown" when nothing has been shown, otherwise list what they have shown.

What counts as evidence about this person:

  - only what the PERSON says, orders, chooses or rejects. What the ASSISTANT offers, describes or assumes
 is scenery, even about a place the person then books.
  - booking a venue is not liking its cuisine or format. If they took it for the distance, the quiet room,
 the price or the availability, the cuisine in the assistant's description is NOT evidenced -- only the
person naming or ordering the cuisine themselves is.
  - a FRIEND, PARTNER or COLLEAGUE's needs are theirs, not this person's. A companion's nut allergy is not
 a nut allergy here.
  - a claim about themselves that their behaviour contradicts. Record what they DO.
  - a CEILING field (formality_ceiling: the highest formality they will go to) is an upper bound. Asking
for or accepting something BELOW it says nothing about where the bound sits; only the highest level they
have actually gone to evidences it.
  - use the level whose words they used, not a neighbour: sushi is not japanese, a market is not a
food_festival.

Output JSON only, no prose outside it.
```

E.2 `memory`: extract call

```
You are reading someone's conversations with a restaurant-recommendation assistant and writing down what
you learn about them, as durable memory items you can consult later without ever re-reading this
conversation.

Write each memory as one atomic, self-contained sentence, with enough of its own evidence baked in that a
later reader who never sees this conversation can judge how strong it is. Prefer "they said directly: ..."
 over a paraphrase that hides whether it was stated, implied, or merely consistent with what they ordered.

What a memory about this person may record:

  - only what the PERSON said, ordered, chose or rejected. What the ASSISTANT offered, described or
assumed is scenery, even about a place the person then booked -- if you record the booking, record the
reason they gave, and mark any cuisine or format as the assistant's description, not theirs.
  - booking a venue is not liking its cuisine or format. Taken for the distance, the quiet room, the price
 or the availability, it evidences nothing about the cuisine; only the person naming or ordering the
cuisine themselves does.
  - a FRIEND, PARTNER or COLLEAGUE's needs are theirs, not this person's. A companion's nut allergy is not
 a nut allergy here.
  - a claim about themselves their behaviour contradicts. Write what they DID, and say so if it conflicts
with something they claimed.
```

```
  - asking for or accepting something BELOW a ceiling (formality_ceiling: the highest formality they will
go to) says nothing about where the ceiling sits. Record the highest level they actually went to, not
every level below it.
  - record the words they used, not a neighbour: sushi is not japanese, a market is not a food_festival.

Skip small talk and anything that reveals nothing about their preferences, habits or constraints.
A conversation with nothing durable in it can produce zero memories -- that is a correct output,
not a failure.

Output JSON only, no prose outside it.
```

E.3 `memory`: integrate call

```
You maintain one person's preference profile from a running memory bank, built from their conversations
with a restaurant-recommendation assistant. You do not see the conversations themselves -- only the memory
 items already on file and the new candidates just written from the latest ones. You are scored on
CALIBRATION, not on coverage. For each tracked field:

  - the memories on file PIN the value   -> give the value. Saying "unknown" scores zero.
  - the memories NARROW it               -> give your best value. "unknown" gets partial credit.
  - no memory touches it                 -> say "unknown". A value here is a FABRICATION and is penalised
EVEN IF IT HAPPENS TO BE CORRECT.

Guessing from a stereotype, a population base rate, or another field earns nothing and costs you whenever
no memory on file supports it.

Seven fields hold LISTS: dietary, allergies, accessibility_needs, cuisines_liked, cuisines_avoided,
event_types_liked, event_types_avoided. On these, an empty list [] is NOT the same as "unknown": it is a
positive claim that the person has NONE, scored as a commitment, so on a field no memory on file
establishes it is a FABRICATION. Two rules:

  - dietary, allergies, accessibility_needs: write [] ONLY once a memory on file says they have none ("no
allergies", "no dietary needs"). Until one does, say "unknown".
  - cuisines_liked, cuisines_avoided, event_types_liked, event_types_avoided: never write [] -- say "
unknown" when no memory shows any, otherwise list what the memories support.

What a memory on file licenses:

  - only what the PERSON said, ordered, chose or rejected. A memory of what the ASSISTANT offered,
described or assumed licenses nothing, even about a place the person booked.
  - booking a venue is not liking its cuisine or format. A memory that they took a venue the assistant
called Mediterranean, for the distance, the quiet room, the price or the availability, does not put
mediterranean in cuisines_liked; only a memory of the person naming or ordering the cuisine does.
  - a FRIEND, PARTNER or COLLEAGUE's needs are theirs, not this person's.
  - a CEILING field (formality_ceiling: the highest formality they will go to) is an upper bound. A memory
 of a request or a booking BELOW it says nothing about where the bound sits; only the highest level on
file evidences it.
  - use the level whose words the memory records, not a neighbour: sushi is not japanese, a market is not
a food_festival.

You also maintain the memory bank itself. For every new candidate below, decide exactly one of:

  - "add"       - genuinely new information; keep it as its own item.
  - "supersede" - it updates or corrects an item already on file; name that item's id in "supersedes" --
the old item is dropped in favour of the new text. A person who says they have stopped eating meat has
changed, and the bank should change with them.
  - "duplicate" - it says nothing the bank does not already have; it is dropped.

Output JSON only, no prose outside it.
```

F Failure-mode catalogue

This appendix backs §7.1 with the procedure, the numbers behind each claim, and the caveats that attach to them.

Runs. The catalogue covers the nine runs of §7, each on every checkpoint of all 50 subjects: `prompted` and `notebook` on all four backbones (Gemini 3.7 Flash, GPT-5.6 Luna, DeepSeek V4 Pro, GLM-5.3 Flash), and `memory` on GPT-5.6 Luna.

Procedure. The catalogue is built in five steps.

(1) *Recover the charged cells.* Every cell the scorer charged (a committed preference scoring below 0.5, a fabrication, or a missed or invented constraint element) is recovered by replaying the cached outputs against the answer key.

(2) *Group them into episodes.* Cells are grouped by (system, backbone, subject, field, element); a new episode opens when checkpoints are not consecutive or the emitted value changes. This gives 807 episodes.

(3) *Read each distinct case once.* Episodes collapse to distinct cases (subject, field, element, emitted value, gold value, first checkpoint); what the transcript supports at a checkpoint does not depend on which system emitted the value, so each case is read once and its verdict applies to every episode in it. Failure readers (Claude Opus 5.5, shown neither the system nor any earlier verdict) read the full served transcript and label each case GENUINE, KEY ARTIFACT, or AMBIGUOUS under the answer key's evidence policy, with a supporting quote, the failure mode, and a flag where the reader doubted the gold.

(4) *Read every non-genuine case again.* The 76 cases not labelled GENUINE were read a second time by independent failure readers who did not see the first verdict. Where the two reads disagreed, a first-read AMBIGUOUS that the second read called GENUINE is taken as GENUINE; the rest were settled from the run data or by the authors.

(5) *Correct the key where it was wrong, and tally.* The reading was first done against the key before its last correction. The failure readers found the key wrong on four fields there: three deal-seeking fields and one avoided cuisine were graded evidenced too late (two) or with no user line behind them (two). Those four are corrected in the released key, the episodes were re-extracted against it, and the three cases that changed were read afresh (all GENUINE). Final mix on the 807 episodes: 742 GENUINE (92%), 60 KEY ARTIFACT (7.4%), 5 AMBIGUOUS. The key artifacts have three causes: 37 episodes (4.6%) are the system recording what the user said under a neighbouring label (a wheat allergy filed as `gluten_free`), 21 (2.6%) are a safe default on a field never raised (`alcohol_free_required` false, an empty list), and 2 (0.2%) are the gold itself being wrong, in two *visit-time* distributions whose hidden histogram disagrees with the visits the transcript shows. These two are left as they are: every distribution is scored against the final card (§6), and re-keying these two to the visits shown would make them the only distributions keyed that way.

Every number in §7.1 is on GENUINE episodes, the "confirmed mistakes" of that section.

Caveats.

(i) *The failure readers are LLMs.* They apply a written policy and are not human annotators. A case the first read called GENUINE was not read again, so most of the GENUINE count (440 of 472 cases) rests on one read.

(ii) *One stateful run.* The stateful mechanism (an unstable re-read, contradictions added rather than superseding, bins starved of data) is read from `memory` on one backbone.

(iii) *Undisclosed rule.* Some turns show an abstract trait (participation, spice) in loose words. The prompts do not tell systems that such a turn supports only that trait, so that every backbone falls for it reflects the prompt as much as the backbone.

Table 5: Confirmed mistakes per subject on the nine runs (all 50 subjects). "—" = run not made. Backbone choice outweighs the scratchpad: the same system spreads by up to 2.3 per subject across backbones (seven-fold), against at most 0.6 for the notes scratchpad.

Backbone	`prompted`	`notebook`	`memory`
Gemini 3.7 Flash	0.38	0.42	—
GPT-5.6 Luna	0.90	0.84	3.14
DeepSeek V4 Pro	2.66	2.04	—
GLM-5.3 Flash	2.10	2.36	—

Early commits. For each fabrication episode, *lead* is the number of checkpoints between the commit and the field's first evidenced checkpoint. Across the single-pass runs the median lead, where finite, is 2–4 checkpoints; 23–45% of fabrications are on a field never evidenced in the subject; the committed value is contained in the final gold in 0–17% of episodes (Gemini 3.7 Flash `prompted` 0/13, GLM-5.3 Flash `notebook` 14/87, GPT-5.6 Luna `prompted` 5/29); and the "lucky" share of all fabrications is 0–16% on every run. Fabrications start early for single-pass systems (median 0.11 of the way through the conversation, 45% at c_0) and later for the stateful one (0.43, 9%).

Ordinal fields. 266 of 323 ordinal errors are commits on a field graded unevidenced. Most common guesses on a field not yet evidenced, pooled over the backbones: *deal-seeking*

Table 6: Share of each backbone's confirmed mistakes by mode (single-pass, both systems pooled, all 50 subjects). *Unevidenced*: a commit on a field not yet evidenced, split by whether the value happens to match the hidden one (*lucky*) or not. *List*: a committed list that is a strict superset or subset of the gold. *Constraint*: an invented or missed element of a hard-constraint field. *Ordinal*: a value two or more levels off. Rows do not sum to 100%; the remainder falls in modes not shown. Shares are rounded separately.

		Unevidenced		List		Constraint		Ordinal
Backbone	n	wrong	lucky	superset	subset	invented	missed	$\geq$2 off
Gemini 3.7 Flash	40	60%	2%	0%	**18%**	0%	10%	0%
GPT-5.6 Luna	87	54%	6%	3%	6%	8%	6%	9%
DeepSeek V4 Pro	235	45%	5%	1%	3%	**30%**	6%	5%
GLM-5.3 Flash	223	**65%**	10%	2%	2%	8%	2%	8%

`notices_value` 87%; *noise* `fine_with_lively` 100%; *design* split between the two middle levels `mild_preference` and `room_matters` (13/9); *formality ceiling* `casual_sit_down` 65% / `counter_service` 30%; *adventurousness* `orders_unfamiliar` 41%. The most common confusions (emitted $\rightarrow$ gold) are `notices_value`$\rightarrow$`plans_around_specials` ($\times$47), `notices_value`$\rightarrow$`looks_for_deals` ($\times$28), `casual_sit_down`$\rightarrow$`white_tablecloth` ($\times$27), and `casual_sit_down`$\rightarrow$`smart_casual` ($\times$17).

Shared triggers. Independent failure readers arrived at the same list; it gives each trigger, a verbatim example, and who falls for it.

Budget cap $\rightarrow$ deal-seeking. "Nothing over fifteen quid if I can help it" (c_0) $\rightarrow$ `notices_value` from seven of the nine runs, on three backbones (Gemini 3.7 Flash is the exception). The single most repeated error line in the catalogue.

Ceiling from the first occasion. "Somewhere we sit up at a counter and order as we go" (c_0) $\rightarrow$ `formality_ceiling=counter_service` from five of the nine runs, on all four backbones; the true ceiling surfaces at c_2 ("I'll be in a jacket"). On Gemini 3.7 Flash and GLM-5.3 Flash `notebook` the prompt's own ceiling clause becomes the reason to commit at c_0 ("highest formality actually gone to is casual sit-down", gold `white_tablecloth`).

Abstract turn read as a concrete event. A turn written to evidence participation or spice is re-read as a concrete event or a nearby trait: "me watching it happen" $\rightarrow$ `live_jazz` avoided by eight of the nine runs; "we went to a wine thing … I was climbing the walls" $\rightarrow$ `wine_tasting` avoided by eight. One of two triggers every backbone falls for on both single-pass systems; the other, a format turned into an event type (below), is the largest single source of invented event types.

Soft preference $\rightarrow$ hard constraint. "Quiet enough to hear myself think" $\rightarrow$ `quiet_room_needed`; "a proper table rather than perching" $\rightarrow$ `seating_required` ($\times$74 across runs, DeepSeek V4 Pro $\times$55; `quiet_room_needed` $\times$22, 16 of them DeepSeek V4 Pro `prompted`, which files the request beside the correct `needs_quiet` after the user has said "no access requirements, stairs are fine"). DeepSeek V4 Pro, GLM-5.3 Flash, GPT-5.6 Luna, and the stateful run; *accessibility needs* is the single hottest field (165 episodes).

A one-off situation read as a trait. "We're twenty minutes out" $\rightarrow$ *travel radius* (five runs, three backbones); "watching the clock" $\rightarrow$ `watches_only`.

A format turned into an event type. A food hall chosen or praised as a format becomes `market`, the most-invented event type ($\times$44: 35 as liked, 9 as avoided; on all nine runs), with the user's own word "market" in the transcript before onset in most of them; the most common guess on a never-raised event field for every backbone. The prompt's "use the level whose words they used" is read as literal word-matching ("The market. I went to one of those" $\rightarrow$ `market`).

Booking $\neq$ liking / assistant label copied. A venue the assistant offered, booked for a side reason (a quiet room, a wheat-free kitchen, its hours), enters *cuisines liked* in the assistant's word ("Javanese and Malay food" $\rightarrow$ `malaysian`). The prompt states this policy to the systems and it largely holds: 31 episodes over nine runs invent a liked cuisine, and the most-invented one (`burgers`, $\times$12) comes from the user's own words, not the assistant's.

Agreement across backbones. For `prompted` (four backbones), 177 distinct (subject, field) cells (a constraint field split by item) fail; 114 (64%) under one backbone, 42 under two, 13 under three, 8 (5%) under all four. `notebook`: 163 cells, 64% / 6%.

Whole-card wipes. The wipe counts of §7.1 are taken from the raw cached outputs of all eight single-pass runs. Nothing in the wiped batches contradicts what was held: eight of DeepSeek V4 Pro `prompted`'s sixteen wipe sessions contain no dining event at all. On that run, 9 held constraints are never recovered (e.g. `alcohol_free_required` held c_5–c_{26}, gone c_{27}–c_{29}). A scratchpad protects a value only when it restates it, which is why DeepSeek V4 Pro wipes less under `notebook` than under `prompted` (6 against 16); a note that only says "no new evidence" does not protect the next checkpoint.

Stateful mechanism. Beside the loss and flicker shares of §7.1, 47% of `memory`'s episodes are over-reach, against 67–76% for the single-pass systems on GPT-5.6 Luna; across all eight single-pass runs, 0–6% of wrong cells flicker. Of seven flickering cells traced through the raw outputs of the two calls, five are the card being re-read differently from an *unchanged* bank (one accessibility item written once yields a card that alternates between two readings thirteen times), and two are the newest candidate winning for one checkpoint and reverting at the next. Across the run, most such flips happen with no relevant change to the bank on either side. Contradictions sit side by side: "no particular diet" and "nothing with wheat" are both on file because the integrate call chose *add*, not *supersede*. Distributions collapse because the dropped data points (§7.1; e.g. "4", "£0–15", "weekend-daytime") leave so few that a single matching point can pile a distribution on one bin for ten checkpoints. Invented accessibility elements are carried, not re-invented: nearly every such card cell has a matching bank item written from the user's literal words, so the over-reading happens when the card is read from the bank.

Lessons for a stateful system. These are drawn from the one stateful run above (caveat (ii)).

- Extract into the schema while the text is still visible.
- Show the integrate call the previous card, and require a cited candidate before a field may change.
- Force a replace-or-reconcile decision on conflict.
- Weight by the number of independent pieces of evidence rather than by position in the prompt.
- Check data points against the bins at extraction time.

Gold the failure readers doubted. Failure readers flagged the gold as doubtful on 112 cases, whatever the verdict; in 88 of them the system is still wrong. Because one disputed value recurs across checkpoints and systems, the 112 come down to 57 questions about the key, in 30 subjects; two of them account for 41 cases (whether a toilet that must be step-free also makes `step_free_access` a need, and whether "I can't have wheat" is an allergy or `gluten_free`). The recurring kinds are a filing choice between neighbouring labels; a hidden value the transcript never surfaces or contradicts (a *visit-time* histogram with a third of its weight on weekday lunches for a user whose every visit served by that checkpoint is at the weekend; a deal-seeking level with no user-side line behind it); a cuisine credited from a booking or the assistant's word rather than the user's; and a value credited at the wrong checkpoint (an end-state value from the first checkpoint, or an element credited before or well after the user voices it). Only the two *visit-time* distributions above were judged wrong by both reads; the rest are one reader's doubt. They are left as keyed for two reasons. Readers saw only cells where a system erred, so editing the key on their doubts alone would move it toward system outputs and never away from them. And most doubts are filing or hidden-value choices that the key settles for every subject by one written policy, which a per-cell edit would break.

G Extended related work

Section 2 places DossierBench against the four lines of work that bound its contribution. This appendix works through the neighbouring benchmarks in the detail the body has no room for. Each is grouped by where it falls relative to the line we draw: whether an unevidenced-but-true claim is charged, and whether abstention is a scored move.

GISTBench in detail. GISTBench [Fostiropoulos et al., 2026] verifies each inferred interest against the engagement signals cited for it, not against a label, and it observes the quadrant we score: the model with the highest recall against self-reported interests has over half its predicted categories unverified. But it cannot charge that *as such*. An unsupported interest lowers its precision whether or not it happens to be true, and the survey truth that would separate a lucky guess from a hallucination sits outside the metric, used only for validation. Its verification is a binary count threshold rather than a graded band; it has no abstention channel (an interest is asserted with citations or simply omitted); and it scores a single snapshot per user, naming longitudinal evaluation as future work.

Benchmarks that reward the move we penalize. StreamProfileBench [Wang et al., 2026b] shares our incremental protocol (profiles inferred over a stream, with earlier batches never re-served) but *rewards* the behavior we penalize: an interest tag absent from the stream so far counts as successful generalization if the user's next batch bears it out. FinPerMA [Wang et al., 2026a] evaluates a persona whose preferences shift along a dated event timeline, but each checkpoint poses a *different* question rather than re-scoring the same field as evidence accrues, and it treats a correct-but-unsupported answer as a legitimate baseline from the model's own knowledge rather than charging for it.

Benchmarks in which honest abstention cannot be expressed. DynamicMem [Xie et al., 2026] keeps a per-checkpoint evidence log, but uses it at construction time to *drop* unevidenced fields from the question set. A true-but-not-yet-knowable field therefore cannot be expressed, and honest abstention scores the same as a wrong guess. MemAudit [Ma et al., 2026], which recovers hidden user state along 31 latent dimensions, likewise scores an explicit `unknown` at zero, below a wrong guess that merely overlaps the topic. In the memory literature the incentive often runs the other way outright: the judge rubric of A-TMA [Shi et al., 2026] states that, for answerable questions, abstaining because the retrieved evidence is insufficient is *wrong*, precisely the move we argue a preference profile must be allowed, and rewarded, to make.

Evidence-sufficiency work outside profiling. FEVER [Thorne et al., 2018] scores a claim against retrieved evidence and has a NOT ENOUGH INFO label; we import that stance into recommendation-side profiling, where the "evidence" is the accumulating dialogue and the "claim" is each field of the profile. Sufficient Context [Joren et al., 2025] builds the same evidence-sufficiency setup but scores a correct answer as correct whether or not the context supported it, and it rejects abstaining on insufficient context precisely because that would forfeit those answers; we take the opposite stance and price that move as a fabrication. GroundEval [Flynt, 2026] states the doctrine in our terms (a response that is true but unreachable under the task's constraints is state-invalid, not correct) but it scores answer correctness and path validity as separate weighted terms, discounting only the path term by the violation rate rather than zeroing the answer, and it offers no abstention channel. Omni-Persona [Oh et al., 2026] does credit abstention directly, but only when the queried persona is entirely absent from a single, fixed, pre-retrieved context. There is no time axis, and the evaluated model is explicitly told to use any partial or indirect answer rather than abstain, which gives up the weak-evidence band we score.

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